

Alisha Karna

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Professional Summary

Experienced Software Developer with **over 7 years** of expertise in **Java, Python, Spring Boot, Microservices, AWS, and Full Stack Development**. Adapt at designing, developing, and maintaining scalable web applications with strong hands-on experience in **Spring Framework, Hibernate, RESTful APIs, Angular, Docker, and Kubernetes**. Skilled in **cloud computing (AWS), DevOps (CI/CD, Jenkins, Docker), and database management (SQL, NoSQL, MongoDB, Cassandra, PostgreSQL)**. Proven ability to collaborate in agile environments, drive high-performance development, and ensure efficient software delivery.

Technical Skills

Programming Languages	Java (8/11/17), TypeScript, JavaScript, Python
Frameworks & Libraries	Spring Boot, Spring MVC, Spring Cloud, Hibernate, NestJS, React, Angular (7/11/12), Node.js, Express.js, Bootstrap
Microservices	Spring Boot, NestJS, Microservices Architecture, REST APIs, GraphQL
Containerization & Orchestration	Docker, Kubernetes, Helm, Google Kubernetes Engine (GKE)
Cloud Services	AWS (EC2, Lambda, S3, RDS, SNS, SQS, Step Functions), GCP (GKE, Pub/Sub, Cloud SQL, Cloud Functions)
Full Stack Development	MERN Stack (MongoDB, Express.js, React, Node.js), NestJS
Event-Driven Architecture	Kafka, Kafka Streams, Apache Flink
Databases	SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB, Elasticsearch), Cassandra
Testing	JUnit, Mockito, Test-Driven Development (TDD)
CI/CD & DevOps	GitLab, Jenkins, Docker, Kubernetes, Terraform, CloudFormation
Messaging & API Documentation	Swagger, REST APIs, GraphQL APIs, Spring Security, OAuth2
Infrastructure Automation & Management	Terraform, CloudFormation
Monitoring & Logging	Splunk, ELK Stack
Version Control	Git, Maven
Development Tools	JIRA, GitLab, IntelliJ IDEA, Eclipse
Agile Methodologies	Scrum, Sprint Planning, Retrospective

Education Details

Master's in Computer Science, University of South Dakota

Professional Experience

U.S. Bank – Minneapolis, MN

Senior Java Full Stack Developer (August 2022 – Present)

- Developed and maintained **microservices-based architecture** using **Spring Boot** and **Spring Cloud**, enabling independent service deployment and scaling.
- Implemented **Test-Driven Development (TDD)** by writing unit tests (JUnit, Mockito) before developing new features, ensuring a high level of code quality and minimizing defects.
- Delivered high-quality, robust code through **iterative development cycles**, ensuring features were released frequently with minimal bugs and aligning with Agile methodologies.
- Designed and implemented **microservices architecture** leveraging **Spring Cloud** for seamless service orchestration and fault tolerance.
- Integrated and orchestrated multiple microservices with **REST APIs** and **GraphQL**, enabling efficient data exchange and enhancing flexibility in client-server communication.
- Designed and implemented **Kafka-based messaging services** for **event-driven architecture**, ensuring efficient data streaming and real-time communication between microservices.
- Built and deployed applications in the **AWS ecosystem**, utilizing services like **EC2, ECS, Lambda, SQS, SNS, S3, and Step Functions** to ensure scalability and high availability.
- Developed **GraphQL APIs** in **Java** to provide optimized, flexible, and scalable data access for front-end applications.
- Engineered **Python-based** data pipelines to process financial datasets, utilizing **Pandas** and **NumPy** for data cleansing, validation, and transformation, ensuring compliance with banking regulations.
- Automated **ETL workflows** with **Python**, integrating **APIs** and cloud solutions to support real-time reporting and fraud detection systems.
- Managed relational and NoSQL databases, optimizing **SQL** queries and implementing database design and performance tuning for **MySQL, DynamoDB, and Elasticsearch**.
- Designed, implemented, and managed **Kubernetes clusters** for **Dockerized** microservices, enabling efficient container orchestration and seamless deployment processes.
- Identified and resolved **system inefficiencies** by implementing **distributed architectures (monolith to microservices migration, async/sync communication optimizations)**.
- Implemented **CI/CD pipelines** with **GitLab**, automating build, testing, and deployment processes to streamline workflows and reduce release cycles.
- Conducted **code reviews** and led mentoring efforts for junior engineers, promoting **technical best practices** and enhancing team productivity.
- Promoted the adoption of **service-oriented architecture (SOA)** and **microservices** to improve system scalability, fault tolerance, and maintainability.
- Collaborated with remote and hybrid teams, including overseas contractors, to ensure successful project delivery and adherence to timelines.
- Led **project planning and integration design discussions**, working with cross-functional teams including product owners, architects, and DevOps engineers.
- Worked with **Terraform** and **CloudFormation** to build **infrastructure as code**, ensuring consistent and repeatable cloud infrastructure deployment.
- Demonstrated leadership in troubleshooting and monitoring **distributed systems**, leveraging tools like **Splunk** to identify and resolve performance issues.
- Designed and optimized **REST APIs with NestJS**, ensuring high availability and low latency.

- Maintained a **Quality Assurance mindset**, ensuring all code was thoroughly tested, including integration and end-to-end testing, before release.
- Conducted **code reviews and architectural discussions**, improving overall team productivity and code quality.
- Actively participated in **Agile Scrum** ceremonies, including sprint planning, stand-ups, and retrospectives, fostering collaboration and continuous improvement.

Tech Stack: *Java 11/17, Spring Boot, Hibernate, AWS (EC2, Lambda, S3, RDS, SNS, SQS, Step Functions), Python, Kubernetes, Docker, Kafka, Jenkins, NestJs Angular 12, MongoDB, MySQL, JIRA, Splunk, Terraform, CloudFormation, GitLab, JUnit, Mockito, GraphQL, DynamoDB, Elasticsearch*

Home Depot – Atlanta, GA

Java Full Stack Developer (April 2020 – July 2022)

- Actively participated in **Agile Scrum** ceremonies, including daily stand-ups, sprint planning, retrospectives, and sprint reviews.
- Contributed to each phase of the **SDLC**, including design, development, testing, deployment, and maintenance, ensuring end-to-end project success.
- Developed **Microservices** for specific business domains, improving service isolation and enhancing maintainability across the system.
- Designed and built **RESTful APIs in Node.js and Express.js**, ensuring efficient communication between microservices and front-end applications.
- Implemented **React components with Redux**, optimizing state management and improving UI responsiveness.
- Utilized **Docker** to containerize **Microservices**, ensuring portability and consistency across development, testing, and production environments.
- Developed **Python** scripts for data extraction and transformation, leveraging **Pandas** and **SQLAlchemy** to analyze sales, inventory, and customer behavior data, enhancing operational efficiency.
- Built automated **ETL** processes using **Python**, connecting cloud storage and **APIs** to streamline product data pipelines and support real-time business insights.
- Designed efficient Docker images with optimized layers to reduce build time and improve deployment speed in containerized applications.
- Orchestrated microservices using **Kubernetes**, automating scaling, load balancing, and deployment across clusters for high availability and fault tolerance.
- Configured **Kubernetes** with **Helm** for managing complex applications and simplifying deployments with reusable templates.
- Integrated **Apache Kafka** and **Flink** to process real-time streaming data, enabling event-driven architecture and ensuring data flow continuity across microservices.
- Developed Kafka producers and consumers for **real-time data streaming**, ensuring high-throughput messaging between services.
- Deployed and managed scalable cloud applications on **GCP** using **Google Kubernetes Engine (GKE)** for container orchestration and **Google Cloud Storage** for secure object storage.
- Developed reusable, functional, and dynamic user interface components using **React**, ensuring consistency and maintainability in web applications.
- Implemented navigation in single-page applications (**SPAs**) using **React Router** for dynamic routing, ensuring a smooth user experience without page reloads.

- Leveraged **Google Pub/Sub** for messaging and event-driven architecture, enabling efficient communication between microservices and distributed systems.
- Utilized **GCP Cloud Functions** to implement serverless compute resources, reducing infrastructure management overhead and scaling automatically based on demand.
- Implemented **GCP Cloud SQL** for database management, ensuring high availability, automated backups, and seamless data replication across regions.
- Gained experience using the **Express.js** framework to create and manage RESTful APIs, route requests, and handle middleware in a scalable manner.
- Automated infrastructure provisioning using **Terraform** with GCP, reducing manual intervention and enhancing consistency in cloud resource management.
- Utilized **GCP Load Balancing** to ensure high availability and fault tolerance, distributing traffic across multiple instances efficiently.
- Created reusable and modular **UI components** using **Angular**, **TypeScript**, **JavaScript**, and **Bootstrap**, improving code maintainability and UI consistency.
- Utilized **JavaScript** for dynamic functionality and enhanced user interactivity on the front end, improving user experience.
- Designed **API documentation** using **Swagger**, making APIs easier to understand and consume for external teams.
- Managed **API security** with **Spring Security** and **OAuth2**, ensuring protected access to the services and resources.
- Utilized **MongoDB as the primary NoSQL database**, optimizing queries and indexing for faster data retrieval.
- Developed **decision trees and architectural blueprints** to guide **integration design choices** between **synchronous vs. asynchronous processing**, **REST APIs vs. event-driven messaging**, and **monolith vs. microservices approaches**.
- Automated deployment pipelines using **Jenkins**, **Docker**, and **Kubernetes**, ensuring consistent and rapid deployment cycles.
- Monitored application logs and performance using **Splunk**, proactively identifying issues and improving system performance.

Tech Stack: *Java 8, Spring Boot, Hibernate, REST APIs, React, Node.js, JavaScript, Python, Kafka, Docker, Kubernetes, Jenkins, GCP (GKE, Pub/Sub), MongoDB, Express.js, PostgreSQL, Splunk, Swagger.*

CVS - Boston, MA

Java Full Stack Developer (January 2018 – March 2020)

- Contributed to the development of features in short, iterative cycles, ensuring the completion of tasks in alignment with **sprint** goals and adjusting work priorities based on feedback.
- Built and maintained backend services using **Java 8**, **Spring Boot**, **Spring MVC**, and **Hibernate ORM**.
- Developed and integrated **REST APIs** for seamless communication between services.
- Designed **Microservices** using **Spring Cloud** for scalable and modular applications.
- Created dynamic front-end components with **Angular 7**, **JavaScript**, **HTML5**, and **CSS3**.
- Implemented **authentication & authorization** using **Spring Security** and **OAuth2**.
- Designed **Python**-based data pipelines to process healthcare and pharmacy data, applying **Pandas** for data validation and cleaning to maintain **HIPAA**-compliant datasets.
- Automated **ETL** processes using **Python**, integrating **APIs** and **Snowflake** to enable real-time data ingestion and support analytics for patient care and supply chain optimization.
- Deployed and managed applications on **AWS EC2**, **S3**, **RDS**, and **Lambda**.

- Wrote and executed basic **SQL** queries (SELECT, INSERT, UPDATE, DELETE) to interact with databases and retrieve required data.
- Used **Docker** for containerization and **Kubernetes** for orchestration and scalability.
- Integrated **Kafka** for **event-driven architecture** and **real-time data streaming**.
- Automated builds and deployments using **Jenkins**, **Maven**, and **Git**.
- Conducted **performance tuning and troubleshooting** to optimize third-party API interactions, reducing latency and enhancing user experience.
- Performed **unit testing** with **JUnit** and **Mockito** and monitored applications with **Splunk** and **ELK Stack**.

Tech Stack: *Agile, Java 8, Spring Boot, Spring MVC, Hibernate, Angular 7, JavaScript, Microservices, REST APIs, , Python, Snowflake, Docker, Kubernetes, AWS, Kafka, Jenkins, GraphQL PostgreSQL, Splunk.*